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Metasurface element, electron tube, and method for producing electron tube

Abstract

A metasurface element includes a support body and a metasurface formed on a surface of the support body. The metasurface includes a metal pattern that is disposed to emit an electron in response to incidence of an electromagnetic wave, and a metal layer that contains an alkali metal and is formed on the metal pattern. The metal layer extends beyond the metal pattern to reach a region on the surface of the support body, the region being not formed with the metal pattern.

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Background/Summary

TECHNICAL FIELD
(1) One aspect of the present disclosure relates to a metasurface element, an electron tube, and a method for producing an electron tube.
BACKGROUND ART
(2) U.S. Unexamined Patent Publication No. **2016**/**0216201** discloses a detection system that detects a terahertz wave. In the detection system, the terahertz wave is incident on a metal layer having a metamaterial structure. When the terahertz wave is incident on the metal layer, electrons are emitted. The emitted electrons react with a surrounding gas to generate light. When the light is detected, the terahertz wave can be detected.
SUMMARY OF INVENTION
Technical Problem
(3) A metasurface element that emits electrons in response to the incidence of an electromagnetic wave is required to have improved sensitivity (electron emission ability). Therefore, an object of one aspect of the present disclosure is to provide a metasurface element and an electron tube having improved sensitivity and a method for producing such an electron tube.
Solution to Problem
(4) According to one aspect of the present disclosure, there is provided a metasurface element including: a support body; and a metasurface formed on a surface of the support body. The metasurface includes a metal pattern that is disposed to emit an electron in response to incidence of an electromagnetic wave, and a metal layer that contains an alkali metal and is formed on the metal pattern. The metal layer extends beyond the metal pattern to reach a region on the surface of the support body, the region being not formed with the metal pattern.
(5) In the metasurface element, the metal layer containing the alkali metal is formed on the metal pattern. Accordingly, the band structure at the surface of the metal pattern can be distorted to reduce the work function, and a potential barrier can be thinned. As a result, the probability of tunneling can be increased, and the sensitivity (electron emission ability) can be improved. In addition, in the metasurface element, the metal layer containing the alkali metal is provided not only on the metal pattern but also in the region on the surface of the support body, the region being not formed with the metal pattern. Accordingly, the resistance value of the surface of the support body can be reduced, and the occurrence of charge-up on the metal pattern can be suppressed. As a result, an event that it becomes difficult for the metasurface to emit an electron due to the charge-up can be suppressed. The sensitivity of the metasurface element is improved by the above configuration.
(6) The metal layer may be a monatomic layer of the alkali metal. In this case, the work function at the surface of the metal pattern can be effectively reduced.
(7) The metal layer may contain an oxide of the alkali metal. In this case, the work function at the surface of the metal pattern can be more effectively reduced.
(8) The alkali metal may be cesium. In this case, the work function at the surface of the metal pattern can be more effectively reduced.
(9) A material of the metal pattern may include gold, platinum, aluminum, graphene, silver, or copper. In this case, the conductivity of the metal pattern can be greatly secured.
(10) The metasurface may include only metal patterns having the same potential as the metal

pattern. In this case, since the insulating property between the metal patterns is not required to be considered, the metal layer can be easily provided in the region on the surface of the support body, the region being not formed with the metal pattern.
(11) The support body may be made of silicon, quartz, sapphire, or zinc selenide. In this case, an electromagnetic wave in a frequency band included in the band from a millimeter wave to infrared light can be transmitted through the support body, and the electromagnetic wave can be incident on the metasurface via the support body.
(12) According to one aspect of the present disclosure, there is provided an electron tube including: the metasurface element; a housing that includes a first window portion through which the electromagnetic wave is transmitted, and accommodates the metasurface element; and an electron multiplier that is disposed inside the housing to multiply an electron emitted from the metasurface element. In the electron tube, the sensitivity is improved due to the above-described reason.
(13) An inside of the housing may be a vacuum. In this case, a residual gas inside the housing can be adsorbed by the metal layer (getter effect), so that the degree of vacuum inside the housing can be increased.
(14) The electron tube according to one aspect of the present disclosure may further include an alkali metal dispenser that is disposed inside the housing to emit the alkali metal when energized. In this case, during production, the alkali metal vaporized by the energization of the alkali metal dispenser can be emitted.
(15) The electron tube according to one aspect of the present disclosure may further include a monitoring member that is disposed inside the housing to emit an electron in response to incidence of a monitoring electromagnetic wave having a wavelength different from a wavelength of the electromagnetic wave. The housing may include a second window portion through which the monitoring electromagnetic wave is transmitted. In this case, during production, the monitoring electromagnetic wave is incident on the monitoring member, and the quantity of the electron emitted from the monitoring member is monitored, so that a change in work function at the surface of the metal pattern can be identified.
(16) According to one aspect of the present disclosure, there is provided a method for producing an electron tube, the method including: a first step of disposing a metasurface element and an alkali supply source inside a housing, in which the metasurface element includes a support body and a metasurface that is formed on a surface of the support body and includes a metal pattern that is disposed to emit an electron in response to incidence of an electromagnetic wave; and a second step of causing the alkali supply source to emit a vaporized alkali metal, and heating the housing to disperse the alkali metal inside the housing to form a metal layer on the metal pattern, the metal layer containing the alkali metal.
(17) In the method for producing an electron tube, the vaporized alkali metal is emitted from the alkali supply source, and the housing is heated to disperse the alkali metal inside the housing to form the metal layer on the metal pattern, the metal layer containing the alkali metal. Since the housing is heated to disperse the alkali metal inside the housing, the metal layer can be reliably formed on the metal pattern. Therefore, according to the method for producing an electron tube, the metal layer containing the alkali metal can be suitably formed on the metal pattern. As a result, the electron tube having improved sensitivity as described above can be produced.
(18) The alkali supply source may be an alkali metal dispenser that emits the alkali metal when energized. In this case, the amount of energization, the time of energization, or the like of the alkali metal dispenser can be adjusted to control the quantity of the alkali metal emitted, so that the metal layer can be formed with good reproducibility.
(19) In the second step, oxygen may be supplied into the housing to form the metal layer on the metal pattern, the metal layer containing an oxide of the alkali metal. In this case, the metal layer containing an oxide of the alkali metal can be formed on the metal pattern.
(20) In the second step, a coil that is disposed outside the housing to surround the housing may be

energized to heat the housing. In this case, the housing can be effectively heated.
(21) In the second step, an inside of the housing may be a vacuum. In this case, a residual gas inside the housing can be adsorbed by the metal layer (getter effect), so that the degree of vacuum inside the housing can be increased.
(22) In the first step, a monitoring member that emits an electron in response to incidence of a monitoring electromagnetic wave having a wavelength different from a wavelength of the electromagnetic wave may be further disposed inside the housing. In the second step, while the monitoring electromagnetic wave is incident on the monitoring member and a quantity of the electron emitted from the monitoring member is monitored, the alkali supply source may be caused to emit the alkali metal. In this case, the quantity of the electron emitted from the monitoring member is monitored, so that a change in work function at the surface of the metal pattern can be identified. As a result, the metal layer can be formed with better reproducibility.
(23) In the first step, the metasurface element and the alkali supply source may be disposed inside the housing such that the metasurface element is separated from the alkali supply source. In the second step, a part of the housing may be heated, the part surrounding the alkali supply source. In this case, since the part of the housing is heated, the part surrounding the alkali supply source, the metal pattern can be suppressed from being heated. As a result, the metal pattern and the metal layer can be suppressed from being alloyed, so that the metal layer can be more suitably formed.
Advantageous Effects of Invention
(24) According to one aspect of the present disclosure, it is possible to provide the metasurface element and the electron tube having improved sensitivity and the method for producing such an electron tube.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a cross-sectional view of an electron tube according to an embodiment.
(2) FIG. **2** is a side view of the electron tube.
(3) FIG. **3** is a partial side view of the electron tube.
(4) FIG. **4** is a plan view of a holding unit.
(5) FIG. **5** is a perspective view of the holding unit.
(6) FIG. **6** is a cross-sectional view of a metasurface element.
(7) FIG. **7** is a view for describing a mode where electrons are emitted in a comparative example.
(8) FIG. **8** is a view for describing a mode where electrons are emitted in the embodiment.
(9) FIG. **9** is a graph showing results of an effect confirmation experiment.
(10) FIG. **10** is a view for describing a method for producing an electron tube.
(11) FIG. **11** is a graph for describing the method for producing an electron tube.
DESCRIPTION OF EMBODIMENTS
(12) Hereinafter, one embodiment of the present invention will be described in detail with reference to the drawings. Incidentally, in the following description, the same reference signs are used for the same or equivalent components and duplicated descriptions will be omitted.
Entire Configuration of Electron Tube
(13) An electron tube **1** illustrated in FIGS. **1** to **3** is configured as a photomultiplier tube (PMT) that outputs an electric signal in response to the incidence of an electromagnetic wave. The electron tube **1** has sensitivity for, for example, an electromagnetic wave in a frequency band from a millimeter wave to infrared light. When the electromagnetic wave is incident into the electron tube **1**, electrons are emitted inside the electron tube **1** and the emitted electrons are multiplied. In the electron tube **1**, the electromagnetic wave is incident on a photoelectric surface, and electrons are emitted from the photoelectric surface due to a tunnel effect (field emission) caused by an intense

electric field.
(14) The electron tube **1** includes a housing **10**, a metasurface element **20**, a holding unit **30**, an electron multiplier **40**, and an electron collector **50**. The metasurface element **20**, the holding unit **30**, the electron multiplier **40**, and the electron collector **50** are disposed inside the housing **10**.
(15) The housing **10** includes a valve **11** and a stein **12**. The housing **10** is formed in, for example, a cylindrical shape. The housing **10** is airtightly sealed by the valve **11** and the stein **12**, and the inside of the housing **10** is held in vacuum. The vacuum in the housing **10** may be an absolute vacuum, or may be filled with a gas of pressure lower than the atmospheric pressure. For example, the internal pressure of the housing **10** may be held at 1×10.sup.−4 to 1×10.sup.−7 Pa. In FIG. **3**, the illustration of the valve **11** is omitted.
(16) A part of the valve **11** forms a window portion (first window portion) **11***a* through which the electromagnetic wave is transmitted. The window portion **11***a* forms a top surface of the housing **10** (one end surface of the housing **10** in an axial direction A). The window portion **11***a* is formed in, for example, a circular flat plate shape perpendicular to the axial direction A. The window portion **11***a* is transmissive to the electromagnetic wave, and transmits at least a part of the frequency band of the electromagnetic wave incident on the window portion **11***a*. The stein **12** forms a bottom surface of the housing **10** (the other end surface of the housing **10** in the axial direction A). The stein **12** is formed in, for example, a circular flat plate shape perpendicular to the axial direction A, and is disposed to overlap the window portion **11***a* when viewed in the axial direction A.
(17) The window portion **11***a* is made of a material depending on the frequency band of the electromagnetic wave incident into the electron tube **1**. For example, the material of the window portion **11***a* includes at least one selected from quartz, silicon, germanium, sapphire, zinc selenide, zinc sulfide, magnesium fluoride, lithium fluoride, barium fluoride, calcium fluoride, magnesium oxide, calcium carbonate, and chalcogenide glass. Accordingly, the electromagnetic wave in a certain band selected from the frequency band from a millimeter wave to infrared light can be guided into the housing **10**. For example, quartz is suitable for transmitting a frequency band of 0.1 THz to 5 THz, silicon is suitable for transmitting a frequency band of 0.04 THz to 11 THz and 46 THz or more, magnesium fluoride is suitable for transmitting a frequency band of 40 THz or more, germanium is suitable for transmitting a frequency band of 13 THz or more, and zinc selenide is suitable for transmitting a frequency band of 14 THz or more.
(18) The housing **10** further includes a plurality of wirings **13** for electrical connection between the inside and the outside of the housing **10**. Each of the wirings **13** is, for example, a lead wire or pin. In this example, each of the wirings **13** is a pin that penetrates through the stein **12** along the axial direction A. Each of the wirings **13** is electrically connected to various members provided inside the housing **10**.
(19) The metasurface element **20** emits electrons in response to the incidence of the electromagnetic wave. The metasurface element **20** includes a support body **21** and a metasurface **22**. The metasurface element **20** is formed in, for example, a substantially rectangular plate shape.
(20) The support body **21** is, for example, a rectangular plate-shaped substrate. The support body **21** has a first surface **21***a* and a second surface **21***b* opposite to the first surface **21***a*. In this example, each of the first surface **21***a* and the second surface **21***b* is a flat surface parallel to the window portion **11***a*. The first surface **21***a* faces the window portion **11***a* at an interval therefrom. The electromagnetic wave transmitted through the window portion **11***a* is incident on the first surface **21***a*. The support body **21** is transmissive to the electromagnetic wave transmitted through the window portion **11***a*, and transmits at least a part of the frequency band of the electromagnetic wave transmitted through the window portion **11***a*.
(21) The support body **21** is made of, for example, silicon, quartz, sapphire, or zinc selenide (ZnSe). The support body **21** is disposed to be separated from the window portion **11***a* and the electron multiplier **40** in the axial direction A. The metasurface **22** is provided on the second surface **21***b* of the support body **21**. The electromagnetic wave transmitted through the window

portion **11***a* and the support body **21** is incident on the metasurface **22**. The metasurface **22** will be described in detail later.
(22) As illustrated in FIGS. **1** to **5**, the holding unit **30** holds the metasurface element **20**. The holding unit **30** includes a bottom wall portion **31** having a annular plate shape and a side wall portion **32** that has a cylindrical shape and extends from an outer edge of the bottom wall portion **31**. A disposition portion **31***a* having a shape corresponding to the external shape of the metasurface element **20** is formed in a central portion of the bottom wall portion **31**, and the metasurface element **20** is disposed inside the disposition portion **31***a*. In this example, the disposition portion **31***a* is formed of an opening formed in the bottom wall portion **31**.
(23) The holding unit **30** is fixed to an inner peripheral surface of the valve **11**, and positions the metasurface element **20** with respect to the housing **10**. For example, the holding unit **30** is positioned with respect to the housing **10** such that an optical axis of the electromagnetic wave passing through the disposition portion **31***a* is parallel to a tube axis (axial direction A) of the housing **10**. The holding unit **30** includes a plurality of (four in this example) terminals **33**. At least one of the plurality of terminals **33** is electrically connected to the wiring **13** via a conduction layer **14**, and a voltage is applied to the metasurface element **20** via the wiring **13**, the conduction layer **14**, and the terminal **33**. The conduction layer **14** is formed on an inner peripheral surface of the valve **11**.
(24) A monitoring member **34** is provided on the bottom wall portion **31** of the holding unit **30**. During production of the electron tube **1**, a monitoring electromagnetic wave having a wavelength different from the wavelength of the electromagnetic wave incident on the metasurface element **20** is incident on the monitoring member **34**. A method for producing the electron tube **1** will be described later.
(25) The monitoring member **34** is disposed adjacent to the metasurface element **20** on a surface on the electron multiplier **40** side (side opposite to the window portion **11***a* and the side wall portion **32**) of the bottom wall portion **31**. The monitoring member **34** is, for example, an electrode in which a metal film is formed on a base material. In this example, the metal film is made of the same metallic material as a metallic material forming a metal pattern **24**, to be described later, of the metasurface **22**. As one example, the material of the metal film is gold, and the material of the base material is beryllium copper (CuBe).
(26) In addition, a through-hole **31***b* is formed in the bottom wall portion **31**. The through-hole **31***b* penetrates through the bottom wall portion **31** along the axial direction A. The monitoring member **34** is disposed such that the monitoring electromagnetic wave passing through the through-hole **31***b* is incident on the monitoring member **34**. The monitoring electromagnetic wave is transmitted through the window portion **11***a* to enter the housing **10**, and then passes through the through-hole **31***b* to be incident on the monitoring member **34**. Namely, the window portion **11***a* also serves as a second window portion through which the monitoring electromagnetic wave is transmitted. The monitoring member **34** is electrically connected to an electrode **23**, to be described later, of the metasurface **22**, and has the same potential as that of the electrode **23**.
(27) The electron multiplier **40** multiplies electrons emitted from the metasurface element **20**. The electron multiplier **40** has an incident surface **40***a* that faces the second surface **21***b* of the support body **21** of the metasurface element **20** at an interval from the second surface **21***b*. The electron multiplier **40** multiplies electrons that are emitted from the metasurface element **20** to be incident on the incident surface **40***a*. The electron multiplier **40** is configured as, for example, a so-called line focus type, and includes a plurality of stages (for example, ten stages) of dynodes. The uppermost dynode includes the incident surface **40***a* described above. Each of the dynodes is electrically connected to the wiring **13**, and a predetermined potential is applied to each of the dynodes via the wiring **13**. The dynode multiplies electrons in accordance with the applied potential.
(28) The electron collector **50** collects the electrons multiplied by the electron multiplier **40**. The

electron collector **50** includes, for example, an anode that is disposed to receive the electrons from the dynode in the last stage of the electron multiplier **40**. The anode is electrically connected to the wiring **13**, and a predetermined potential is applied to the anode via the wiring **13**. The anode captures the electrons multiplied by the dynodes. The electron collector **50** may include a diode (avalanche diode or the like) instead of the anode.
(29) The electron tube **1** further includes an alkali metal dispenser (AMD) **60** disposed inside the housing **10**. The alkali metal dispenser **60** is an alkali supply source that emits an alkali metal when energized. The alkali metal dispenser **60** is used during production of the electron tube **1** as will be described later. The method for producing the electron tube **1** will be described later. The alkali metal dispenser **60** is formed in, for example, a bar shape, and extends along the axial direction A. The alkali metal dispenser **60** is disposed in a space between the electron multiplier **40** and the inner surface of the housing **10** (inner peripheral surface of the valve **11**), and extends toward the stein **12**. The alkali metal dispenser **60** is separated from the metasurface element **20** in the axial direction A.
(30) The alkali metal dispenser **60** includes an accommodation member **61** made of a metallic material, and the alkali metal is accommodated in the accommodation member **61**. As one example, the material of the accommodation member **61** is nickel and chromium, and the alkali metal accommodated is cesium. The accommodation member **61** is electrically connected to the wiring **13**. When current flows through the accommodation member **61** via the wiring **13**, the accommodation member **61** is resistance heated, and the alkali metal vaporized is emitted from the accommodation member **61**. The amount of energization or the time of energization of the accommodation member **61** can be adjusted to control the quantity of the alkali metal emitted.
(31) An emission port through which the alkali metal is emitted is formed in the accommodation member **61**. The emission port faces the inner peripheral surface of the valve **11**. For this reason, the alkali metal is emitted from the accommodation member **61** toward the inner peripheral surface of the valve **11**. Accordingly, in a production process to be described later, the alkali metal is suppressed from being adsorbed on the electron multiplier **40** or the electron collector **50**; and thereby, the probability of adsorption of the alkali metal on the metal pattern **24** can be increased.
Configuration of Metasurface Element
(32) As illustrated in FIGS. **4** to **6**, the metasurface element **20** includes the support body **21** described above and the metasurface **22** that is formed on the second surface **21***b* of the support body **21**. The metasurface **22** includes the electrode **23**, the metal pattern **24**, and a metal layer **25**. The metasurface **22** faces the window portion **11***a* on one side in the axial direction A with the support body **21** disposed therebetween, and faces the incident surface **40***a* of the electron multiplier **40** on the other side in the axial direction A.
(33) The electrode **23** is formed on the second surface **21***b* of the support body **21**, and has a substantially rectangular frame shape in plan view (when viewed in a direction perpendicular to the second surface **21***b* (direction parallel to the axial direction A)). A voltage is applied to the electrode **23** via the terminals **33**.
(34) The metal pattern **24** is formed in layer shape on the second surface **21***b* so as to be located inside the electrode **23** in plan view. The metal pattern **24** is disposed to emit electrons in response to the incidence of the electromagnetic wave. As a disposition of the metal pattern **24**, for example, a pattern described in U.S. Unexamined Patent Publication No. 2016/0216201 may be used. In this example, the metal pattern **24** includes a plurality of conduction wires **24***a* that are disposed to emit electrons when an electromagnetic wave (for example, terahertz wave) in a certain band selected from the frequency band from a millimeter wave to infrared light is incident. For example, the plurality of conduction wires **24***a* have the same shape, and are arranged along a predetermined direction. Each of the conduction wires **24***a* is electrically connected to the electrode **23**. Accordingly, the plurality of conduction wires **24***a* have the same potential. Namely, in this example, the metal pattern **24** has only metal patterns having the same potential.

(35) The metal pattern **24** includes an oxide layer formed on the second surface **21***b* of the support body **21**, and a metal layer formed on the oxide layer. The oxide layer includes a silicon dioxide layer formed on the second surface **21***b*, and a titanium oxide layer formed on the silicon dioxide layer. As one example, the thickness of the silicon dioxide layer is approximately 1 μm, and the thickness of the titanium oxide layer is approximately 10 nm. As one example, the material of the metal layer is gold, and the thickness of the metal layer is approximately 200 nm. The material of the metal pattern **24** does not contain the alkali metal. The electrode **23** has, for example, the same layer structure as that of the metal pattern **24**.
(36) The metal layer **25** is formed on the electrode **23** and the metal pattern **24** to cover the electrode **23** and the metal pattern **24**. The metal layer **25** extends beyond the metal pattern **24** to reach a region Ron the second surface **21***b* of the support body **21**, the region R being not formed with the electrode **23** and the metal pattern **24**. Namely, the metal layer **25** is provided to straddle the metal pattern **24** and the region R on the second surface **21***b*. In this example, the metal layer **25** is provided over the entire surface of the electrode **23**, the entire surface of the metal pattern **24**, and the entirety of the region R. In other words, the metal layer **25** is provided over the entirety of a region on the second surface **21***b*, the region being located inside an outer edge of the electrode **23**.
(37) The material of the metal layer **25** contains the alkali metal. The metal layer **25** is, for example, an oxide layer of the alkali metal. In this example, the alkali metal is cesium, and the metal layer **25** is a cesium oxide layer. The thickness of the metal layer **25** is, for example, approximately several angstroms. The work function of the material of the metal layer **25** is lower than the work function of the metallic material (gold in this example) forming the metal pattern **24**.
Functions and Effects
(38) In the metasurface element **20**, the metal layer **25** containing the alkali metal is formed on the metal pattern **24**. Accordingly, the band structure at the surface of the metal pattern **24** can be distorted to reduce the work function, and a potential barrier can be thinned. As a result, the probability of tunneling can be increased, and the sensitivity (electron emission ability) can be improved. In addition, in the metasurface element **20**, the metal layer **25** containing the alkali metal is provided not only on the metal pattern **24** but also in the region R on the second surface **21***b* of the support body **21**, the region R being not formed with the metal pattern **24**. Accordingly, the resistance value of the second surface **21***b* of the support body **21** can be reduced, and the occurrence of charge-up on the metal pattern **24** can be suppressed. As a result, an event that it becomes difficult for the metasurface **22** to emit electrons due to the charge-up can be suppressed. The sensitivity of the metasurface element **20** is improved by the above configuration. The electron tube **1** including the metasurface element **20** described above can be used in various fields as an uncooled and highly sensitive detector capable of detection in a wide range from a mid-infrared region to a terahertz region. The charge-up is a phenomenon where the emitted electrons collide with the region R, in which the metal pattern **24** is not formed, to cause the region R to be positively or negatively charged. When the charge-up occurs, it is difficult for the metasurface **22** to emit electrons.
(39) The above points will be further described with reference to the FIGS. **7** and **8**. FIG. **7** is a view for describing a mode where electrons are emitted from the metasurface **22** in a comparative example, and FIG. **8** is a view for describing a mode where electrons are emitted from the metasurface **22** in the embodiment. The comparative example corresponds to a configuration where the metal layer **25** is not provided on the metasurface **22** of the embodiment. In FIGS. **7** and **8**, the vertical axis represents potential energy U, and the horizontal axis represents a distance x from the metal pattern **24**.
(40) In both of the comparative example and the embodiment, when an electromagnetic wave is incident on the metasurface **22**, as indicated by an arrow and a broken line B, the potential in the vicinity of the metal pattern **24** is inclined. Accordingly, as compared to a state where no voltage is applied, the quantity of electrons e passing through the potential barrier due to the tunnel effect is

further increased.
(41) As illustrated in FIGS. **7** and **8**, when the metal layer **25** containing the alkali metal is formed on the metal pattern **24** (FIG. **8**), a potential barrier PB is expected to be thinner as compared to when the metal layer **25** is not formed on the metal pattern **24** (FIG. **7**). It is considered that the reason is, as illustrated in FIG. **8**, that the band structure at the surface of the metal pattern **24** is distorted by the metal layer **25** to reduce the work function. Namely, it is considered that when the metal layer **25** is provided, the work function at the surface of the metal pattern **24** is smaller as compared to when the metal layer **25** is not provided. As a result, the probability of tunneling can be increased; and thereby, an effect that the sensitivity (electron emission ability) can be improved is expected.
(42) In addition, in the metasurface element **20** of the embodiment, primary electrons emitted from the metasurface **22** collide with the metal pattern **24** or the like around the metasurface **22** to cause secondary electrons to be emitted. It is considered that since the metal layer **25** is formed on the metal pattern **24**, not only the quantity of the emitted primary electrons but also the quantity of the emitted secondary electrons is increased.
(43) FIG. **9** is a graph showing results of an effect confirmation experiment. In FIG. **9**, the horizontal axis represents the intensity of an incident electromagnetic wave, and the vertical axis represents the magnitude of an output voltage. In the experiment, five electron tubes corresponding to the electron tube **1** of the embodiment were prepared as examples, and the sensitivity of each of the electron tubes was measured. Data S**1** to S**5** represent measurement data of the examples. An alternate long and two short dashes line L represents the maximum value of the sensitivity in the comparative example. The comparative example corresponds to the configuration where the metal layer **25** is not provided on the metasurface **22** of the embodiment.
(44) As shown in FIG. **9**, the sensitivity of any one of the examples was higher than that of the comparative example. In addition, in the examples, the sensitivity up to a maximum of **36** times that of the comparative example could be realized. Specifically, the threshold value of the comparative example was 20 kV/cm, whereas the threshold value of the example of the data S**1** was 0.553 kV/cm.
(45) Subsequently, the effect of the metasurface element **20** will be described. In the metasurface element **20**, the metal layer **25** contains an oxide of the alkali metal. The alkali metal forming the metal layer **25** is cesium. Accordingly, the work function at the surface of the metal pattern **24** can be more effectively reduced.
(46) The material of the metal pattern **24** contains gold. Accordingly, the conductivity of the metal pattern **24** can be greatly secured.
(47) The metasurface **22** includes only the metal patterns **24** having the same potential as the metal pattern. Accordingly, since the insulating property between the metal patterns **24** is not required to be considered, the metal layer **25** can be easily provided in the region R on the second surface **21***b* of the support body **21**, the region R being not formed with the metal pattern **24**.
(48) The support body **21** is made of silicon, quartz, sapphire, or zinc selenide. Accordingly, an electromagnetic wave in a frequency band included in the band from a millimeter wave to infrared light can be transmitted through the support body **21**, and the electromagnetic wave can be incident on the metasurface **22** via the support body **21**.
(49) In the electron tube **1**, the inside of the housing **10** is a vacuum. Accordingly, a residual gas inside the housing **10** can be adsorbed by the metal layer **25** (getter effect), so that the degree of vacuum inside the housing **10** can be increased.
(50) The electron tube **1** includes the alkali metal dispenser **60** that is disposed inside the housing **10** to emit the alkali metal when energized. Accordingly, during production, the alkali metal vaporized by the energization of the alkali metal dispenser **60** can be emitted.
(51) The electron tube **1** includes the monitoring member **34** that emits electrons in response to the incidence of a monitoring electromagnetic wave, and the housing **10** includes the window portion

11*a* through which the monitoring electromagnetic wave is transmitted. Accordingly, during production, the monitoring electromagnetic wave is incident on the monitoring member **34**, and the quantity of electrons emitted from the monitoring member **34** is monitored, so that a change in work function at the surface of the metal pattern **24** can be identified.

Method for Producing Electron Tube

(52) One example of the method for producing the electron tube **1** described above will be described. First, as illustrated in FIG. **10**, the metasurface element **20** and the alkali metal dispenser **60** are disposed inside the housing **10** (first step). The metasurface element **20** is a metasurface element before the metal layer **25** is formed on the metal pattern **24**. In the first step, the holding unit **30**, the electron multiplier **40**, the electron collector **50**, and the like that are other components of the electron tube **1** are also disposed inside the housing **10**. The monitoring member **34** provided to the holding unit **30** is also disposed inside the housing **10**. In the first step, the metasurface element **20** and the alkali metal dispenser **60** are disposed inside the housing **10** such that the metasurface element **20** is separated from the alkali metal dispenser **60** in the axial direction A.

(53) Subsequently, the alkali metal vaporized by the energization of the alkali metal dispenser **60** is emitted, and the housing **10** is heated to disperse the alkali metal inside the housing **10** to form the metal layer **25** on the metal pattern **24**, the metal layer **25** containing the alkali metal (second step). Before the second step is started, the housing **10** is sealed, and the inside of the housing **10** is maintained in vacuum.

(54) As illustrated in FIG. **10**, in the second step, a coil **70** disposed outside the housing **10** to surround the housing **10** is energized to heat a part **10***a* of the housing **10**. The part **10***a* of the housing **10** is a part of the housing **10** in the axial direction A, and is a portion that surrounds the alkali metal dispenser **60**. The coil **70** is spirally wound a plurality of times to surround the part **10***a* of the housing **10**. When the coil **70** is energized, only the part **10***a* of the housing **10** is heated. As a result, the metasurface element **20** can be suppressed from being heated. In addition, in the second step, oxygen is supplied into the housing **10** to form the metal layer **25** on the metal pattern **24**, the metal layer **25** containing an oxide of the alkali metal. For example, oxygen is supplied from an oxygen cylinder provided in an exhaust system and connected to an internal space of the housing **10**. The oxygen cylinder is provided with a valve, and the quantity of supply of oxygen can be finely adjusted by the valve.

(55) In addition, in the second step, while a monitoring electromagnetic wave is incident on the monitoring member **34**, and the quantity of electrons emitted from the monitoring member **34** is monitored, the alkali metal dispenser **60** is energized. The monitoring electromagnetic wave is an electromagnetic wave having a wavelength different from the wavelength of an electromagnetic wave incident on the metasurface element **20**. In this example, as described above, a metal film made of gold is formed on the surface of the monitoring member **34**. The work function of gold is 5.47 eV which corresponds to 227 nm when converted to wavelength. Therefore, for example, when the monitoring member **34** is irradiated with ultraviolet light having a wavelength shorter than 227 nm as a monitoring electromagnetic wave, photoelectrons are emitted from the monitoring member **34** due to the photoelectric effect. For example, the electrons emitted from the monitoring member **34** are multiplied by the electron multiplier **40** to be collected by the electron collector **50**. Alternatively, the electron tube **1** may include components which multiply and collect the electrons emitted from the monitoring member **34**, separately from the electron multiplier **40** and/or the electron collector **50**.

(56) The quantity of the electrons emitted from the monitoring member **34** corresponds to the quantity of electrons emitted from the metal pattern **24**. The reason is that the alkali metal emitted from the alkali metal dispenser **60** is deposited not only on the metal pattern **24** but also on the monitoring member **34**. For this reason, when the work function at the surface of the monitoring member **34** is reduced, it can be regarded that the work function at the surface of the metal pattern **24** is also reduced. In other words, the monitoring member **34** can be regarded as a virtual

metasurface. For example, a D2 (deuterium) lamp can be used as a light source that outputs a monitoring electromagnetic wave. In FIG. **10**, a light source **80** is illustrated.
(57) The second step will be further described with reference to FIG. **11**. In FIG. **11**, the horizontal axis represents the elapsed time from the start of energization, and the vertical axis represents the magnitude of an output current. As shown in FIG. **11**, in a first section T**1** immediately after the start of energization, a predetermined current is applied to the alkali metal dispenser **60** to introduce the alkali metal into the housing **10**. Subsequently, in a second section T**2**, the predetermined current is applied to the alkali metal dispenser **60**, and oxygen is supplied into the housing **10** at a predetermined pressure to introduce the alkali metal and the oxygen into the housing **10**. The metal layer **25** can be formed on the metal pattern **24** by such a step. Incidentally, the quantity of current applied to the alkali metal dispenser **60** and/or the supply pressure of oxygen may be changed in accordance with the elapsed time.
(58) As described above, in the method for producing the electron tube **1**, the alkali metal vaporized in the alkali metal dispenser **60** (alkali supply source) is emitted, and the housing **10** is heated to disperse the alkali metal inside the housing **10** to form the metal layer **25** on the metal pattern **24**, the metal layer **25** containing the alkali metal. Since the housing **10** is heated to disperse the alkali metal inside the housing **10**, the metal layer **25** can be reliably formed on the metal pattern **24**. Namely, if the housing **10** is not heated, there is a possibility that the alkali metal emitted stays in the vicinity of the alkali metal dispenser **60**; however, since the housing **10** is heated, such a situation can be suppressed. Therefore, according to the method for producing the electron tube **1**, the metal layer **25** containing the alkali metal can be suitably formed on the metal pattern **24**. Namely, the electron tube **1** of which the sensitivity is improved due to the above-described reason can be produced.
(59) The alkali metal dispenser **60** is used as an alkali supply source. Accordingly, the amount of energization, the time of energization, or the like of the alkali metal dispenser **60** can be adjusted to control the quantity of the alkali metal emitted, so that the metal layer **25** can be formed with good reproducibility. In addition, the control of the quantity of the alkali metal emitted, which is performed by using the alkali metal dispenser **60**, is unlikely to cause an operator-dependent variation, and can be easily automated.
(60) In the second step, oxygen is supplied into the housing **10** to form the metal layer **25** on the metal pattern **24**, the metal layer **25** containing an oxide of the alkali metal. Accordingly, the metal layer **25** containing an oxide of the alkali metal can be formed on the metal pattern **24**.
(61) In the second step, the coil **70** disposed outside the housing **10** to surround the housing **10** is energized to heat the housing **10**. Accordingly, the housing **10** can be effectively heated.
(62) In the second step, the inside of the housing **10** is a vacuum. Accordingly, a residual gas inside the housing **10** can be adsorbed by the metal layer **25** (getter effect), so that the degree of vacuum inside the housing **10** can be increased. In addition, the alkali metal can be handled in vacuum.
(63) In the second step, while a monitoring electromagnetic wave is incident on the monitoring member **34**, and the quantity of electrons emitted from the monitoring member **34** is monitored, the alkali metal dispenser **60** is energized. Accordingly, the quantity of electrons emitted from the monitoring member **34** is monitored, so that a change in work function at the surface of the metal pattern **24** can be identified. As a result, the metal layer **25** can be formed with better reproducibility. The monitoring using the monitoring member **34** describe above is effective when it is difficult to use a light source that outputs an electromagnetic wave to which the electron tube **1** has sensitivity (for example, a terahertz light source that outputs a terahertz wave). When this application is filed, the terahertz light source is very expensive and is not easy to use.
(64) In the first step, the metasurface element **20** and the alkali metal dispenser **60** are disposed inside the housing **10** such that the metasurface element **20** is separated from the alkali metal dispenser **60**, and in the second step, the part **10*a*** of the housing **10** is heated, the part **10*a*** surrounding the alkali metal dispenser **60**. Accordingly, since the part **10*a*** of the housing **10** is

heated, the part **10***a* surrounding the alkali metal dispenser **60**, the metal pattern **24** can be suppressed from being heated. Namely, the activity at room temperature can be realized. As a result, the metal pattern **24** and the metal layer **25** can be suppressed from being alloyed, so that the metal layer **25** can be more suitably formed.
(65) For example, in the method for producing the electron tube **1** described above, the temperature of the metasurface **22** is maintained at approximately 100° C. or less. The temperature of the metasurface **22** may be maintained at approximately 60° C. or less, or may be maintained at approximately 25° C. Meanwhile, the part **10***a* of the housing **10** is heated to approximately 200° C. by the coil **70**. Namely, in the second step, the temperature of the metasurface **22** is maintained at a temperature lower than the temperature of the part **10***a* of the housing **10**.
(66) In the method for producing the electron tube **1** described above, the alkali metal is supplied by using the alkali metal dispenser **60**; however, as another method, a method using an ampoule or a pellet can be considered. The former is a method by which an operator uses an ampoule tube to introduce the alkali metal into the housing **10** and pulls out the ampoule tube from the housing **10** after the introduction is completed. This method has a drawback that the reproducibility in forming the metal layer **25** is lower and the temperature of the metasurface **22** is more likely to become high as compared to the production method of the embodiment.
(67) The method using a pellet is a method by which a mass (pellet) of the alkali metal disposed inside the housing **10** is heated at a high frequency to supply the alkali metal. This method has a drawback that the temperature of the metasurface **22** is more likely to become high and the quantity of supply of the alkali metal is more difficult to control as compared to the production method of the embodiment. The production method of the embodiment has an advantage that the temperature of the metasurface **22** is less likely to become high, the quantity of supply of the alkali metal is easier to control, and the reproducibility in forming the metal layer **25** is higher as compared to this method.
(68) One embodiment of the present disclosure has been described above; however, the present disclosure is not limited to the embodiment. The materials and the shapes of each configuration are not limited to the materials and the shapes described above, and various materials and shapes can be adopted. For example, the electrode **23** is not limited to having a rectangular frame shape, and may have any shape such as a rectangular shape. The metal pattern **24** may have any disposition as long as the metal pattern **24** can emit electrons in response to the incidence of an electromagnetic wave.
(69) In the embodiment, the metal layer **25** is formed on the electrode **23** and the metal pattern **24**; however, the metal layer **25** may be formed only on the metal pattern **24**. In the embodiment, the metal layer **25** is provided over the entirety of the surface of the metal pattern **24**; however, the metal layer **25** may be formed on at least a part of the metal pattern **24**. In the embodiment, the metal pattern **24** includes only metal patterns having the same potential; however, the metal pattern **24** may include two or more metal patterns to which different potentials are applied.
(70) The housing **10** may include a second window portion through which a monitoring electromagnetic wave is transmitted, separately from the window portion **11***a*. The metasurface **22** may be formed on the first surface **21***a* of the support body **21**. In this case, the support body **21** may not necessarily be transmissive to an electromagnetic wave transmitted through the window portion **11***a*. The electron multiplier **40** and the electron collector **50** are not limited to the above-described configurations, and may have any configurations. The monitoring member **34** and/or the alkali metal dispenser **60** may not be disposed inside the housing **10**.
(71) The metal layer **25** may be a monatomic layer of the alkali metal. In this case, the work function at the surface of the metal pattern **24** can be effectively reduced. For example, the metal layer **25** may be a monatomic layer of cesium. The material of the metal pattern **24** is not limited to gold, and may be platinum, aluminum, graphene, silver, or copper. Also in this case, the conductivity of the metal pattern **24** can be greatly secured. The material of the metal layer **25** may

be an alkali metal other than cesium. Another alkali supply source may be used instead of the alkali metal dispenser **60**.
(72) In the embodiment, the metasurface element **20** is configured as a transmission type in which an electromagnetic wave transmitted through the support body **21** is incident on the metasurface **22**; however, the metasurface element **20** may be configured as a reflection type in which an electromagnetic wave is incident on the metasurface **22** from a side opposite to the support body **21**, and electrons are emitted from the metasurface **22** to the side opposite to the support body **21**. In this case, the support body **21** may not necessarily be made of a material that is transmissive to an electromagnetic wave.

Claims

1. A metasurface element comprising: a support body; and a metasurface formed on a surface of the support body, wherein the metasurface includes a metal pattern that is disposed to emit an electron, due to field emission, in response to incidence of an electromagnetic wave, and a metal layer that contains an alkali metal and is formed on the metal pattern, and the metal layer extends beyond the metal pattern to reach a region on the surface of the support body, the region being not formed with the metal pattern.
2. The metasurface element according to claim 1, wherein the metal layer is a monatomic layer of the alkali metal.
3. The metasurface element according to claim 1, wherein the metal layer contains an oxide of the alkali metal.
4. The metasurface element according to claim 1, wherein the alkali metal is cesium.
5. The metasurface element according to claim 1, wherein a material of the metal pattern includes gold, platinum, aluminum, graphene, silver, or copper.
6. The metasurface element according to claim 1, wherein the metasurface includes only metal patterns having the same potential as the metal pattern.
7. The metasurface element according to claim 1, wherein the support body is made of silicon, quartz, sapphire, or zinc selenide.
8. An electron tube comprising: the metasurface element according to claim 1; a housing including a first window portion through which the electromagnetic wave is transmitted and accommodating the metasurface element; and an electron multiplier disposed inside the housing to multiply an electron emitted from the metasurface element.
9. The electron tube according to claim 8, wherein an inside of the housing is a vacuum.
10. The electron tube according to claim 8, further comprising: an alkali metal dispenser disposed inside the housing to emit the alkali metal when energized.
11. The electron tube according to claim 8, further comprising: a monitoring member disposed inside the housing to emit an electron in response to incidence of a monitoring electromagnetic wave having a wavelength different from a wavelength of the electromagnetic wave, wherein the housing includes a second window portion through which the monitoring electromagnetic wave is transmitted.
12. The metasurface element according to claim 1, wherein the metal pattern is disposed to emit an electron, due to field emission, in response to incidence of a terahertz wave.
13. A method for producing an electron tube, the method comprising: a first step of disposing a metasurface element and an alkali supply source inside a housing, wherein the metasurface element includes a support body and a metasurface that is formed on a surface of the support body and includes a metal pattern disposed to emit an electron, due to field emission, in response to incidence of an electromagnetic wave; and a second step of causing the alkali supply source to emit a vaporized alkali metal, and heating the housing to disperse the alkali metal inside the housing to form a metal layer on the metal pattern, the metal layer containing the alkali metal.

14. The method for producing an electron tube according to claim 13, wherein the alkali supply source is an alkali metal dispenser that emits the alkali metal when energized.

15. The method for producing an electron tube according to claim 13, wherein in the second step, oxygen is supplied into the housing to form the metal layer on the metal pattern, the metal layer containing an oxide of the alkali metal.

16. The method for producing an electron tube according to claim 13, wherein in the second step, a coil disposed outside the housing to surround the housing is energized to heat the housing.

17. The method for producing an electron tube according to claim 13, wherein in the second step, an inside of the housing is a vacuum.

18. The method for producing an electron tube according to claim 13, wherein in the first step, a monitoring member that emits an electron in response to incidence of a monitoring electromagnetic wave having a wavelength different from a wavelength of the electromagnetic wave is further disposed inside the housing, and in the second step, while the monitoring electromagnetic wave is incident on the monitoring member and a quantity of the electron emitted from the monitoring member is monitored, the alkali supply source is caused to emit the alkali metal.

19. The method for producing an electron tube according to claim 13, wherein in the first step, the metasurface element and the alkali supply source are disposed inside the housing such that the metasurface element is separated from the alkali supply source, and in the second step, a part of the housing is heated, the part surrounding the alkali supply source.